Functional Annotation of *betC* (Choline-O-sulfatase, O69787) in *Sinorhizobium meliloti* 1021

Gene/Protein Identity (verified)

- **UniProt:** O69787 — Choline-sulfatase, **EC 3.1.6.6**
- **Gene:** *betC* (ordered locus **R00949**; ORF **SMc00127**)
- **Organism:** *Rhizobium meliloti* strain 1021 (*Ensifer / Sinorhizobium meliloti*)
- **Family:** Sulfatase (formylglycine-dependent) family; domains: Sulfatase_N (IPR000917), Sulfatase_CS (IPR024607), Choline-sulfatase (IPR017785), Choline_sulf_C_dom (IPR025863), Alkaline-phosphatase-core superfamily (IPR017850)

Identity confirmation: The gene symbol *betC*, the organism, the EC number (3.1.6.6), and the sulfatase family/domain architecture are mutually consistent and are directly supported by primary literature on the *S. meliloti betICBA* operon (Osterås et al. 1998, P 9736747; Mandon et al. 2003, P 12906115). This is the correct target; literature matches precisely. (Note: a *betC* with the same name and function exists in *Pseudomonas putida* and *Bacillus subtilis*; these are genuine orthologs/analogs, not mis-identifications — see "Comparative context.")

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1. Summary (answer to the research question)

BetC is a choline-O-sulfatase (EC 3.1.6.6), a cytoplasmic, formylglycine-dependent sulfatase that hydrolyzes the sulfate-ester bond of choline-O-sulfate to yield free choline plus inorganic sulfate (it also acts, more slowly, on phosphorylcholine). It is the first (entry) enzyme of the *betICBA* pathway in *S. meliloti*: the choline it releases is oxidized by choline

dehydrogenase (BetA) and betaine-aldehyde dehydrogenase (BetB) to the osmoprotectant **glycine betaine**. BetC therefore routes plant/fungal-derived choline-O-sulfate into (i) osmoprotection via glycine betaine and (ii) nutrient catabolism, enabling *S. meliloti* to use choline-O-sulfate as a sole carbon, nitrogen, **and** sulfur source. It functions in the bacterial cytoplasm and is expressed symbiotically within infection threads and nodule bacteroids.

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2. Primary Function — Reaction and Substrate Specificity

Catalyzed reaction (EC 3.1.6.6):

> choline-O-sulfate + H_2O → choline + sulfate

Osterås et al. (1998) identified *betC* as "encoding a choline sulfatase catalyzing the conversion of choline-O-sulfate and, at a lower rate, phosphorylcholine, into choline" (P 9736747).
Thus:

- **Primary substrate:** choline-O-sulfate (the physiologically relevant substrate).
- **Secondary substrate:** phosphorylcholine (hydrolyzed at a lower rate), indicating the active site accommodates the choline moiety with different anionic leaving groups (sulfate vs. phosphate ester).
- **Products:** free choline + inorganic sulfate (or phosphate from phosphorylcholine).

Genetic assignment of the activity: Choline sulfatase activity was "absent from *betC* but not from *betB* mutants" (P 9736747), unambiguously assigning the enzymatic activity to the *betC* gene product rather than to other *bet* enzymes. The activity is enzymatically **inducible** — induced "indifferently by choline or choline-O-sulfate," in parallel with the other pathway enzymes (P 9736747).

Catalytic mechanism (by family/structural inference)

BetC is a member of the **formylglycine (FGly)-dependent sulfatase family** (Sulfatase_N / Sulfatase_CS domains; alkaline-phosphatase-core superfamily). Although BetC itself has not been crystallized, the mechanism is firmly established for the family from high-resolution structures of close bacterial and human relatives:

- A strictly conserved active-site cysteine (within the sulfatase C-X-P-X-R signature) is **post-translationally oxidized to Cα-formylglycine (FGly)**; this modification "is strictly conserved in prokaryotic and eukaryotic sulfatases" (Ghosh 2007, P 17558559).
- The 1.3 Å structure of *Pseudomonas aeruginosa* arylsulfatase (PAS) showed that "the resting state of the key catalytic residue in sulfatases is a formylglycine hydrate" and established "a mechanism for sulfate ester cleavage involving an aldehyde hydrate as the functional group that initiates the reaction through a nucleophilic attack on the sulfur atom in the substrate" (Boltes et al. 2001, P 11435113).
- Catalysis requires a divalent metal cation (Ca^{2+} in PAS) that positions the substrate sulfate and stabilizes the transition state (P 11435113).

Mechanistic inference for BetC: the FGly-hydrate gem-diol performs nucleophilic attack on the sulfur of the choline sulfate ester, transiently forming a covalent sulfo-enzyme intermediate; elimination releases choline (the alcohol), and hydrolysis regenerates the enzyme while liberating inorganic sulfate. This explains both the measured choline-sulfatase activity and the ability of the enzyme to also process the analogous phosphate ester (phosphorylcholine), albeit less efficiently.

BetC-specific sequence evidence (this work): Direct analysis of the 512-residue O69787 sequence confirms the mechanism is not merely homology-inferred. The conserved sulfatase FGly-directing signature **C-X-P-X-R is present as C54-A-P-A-R** (residues 54–58). UniProt feature annotation marks **residue 54 as the catalytic active-site nucleophile, post-translationally modified to 3-oxoalanine (Cα-formylglycine) from cysteine**, with a second active-site residue at position 104 and metal/substrate-binding residues at positions 14, 15, 296 and 297 (the divalent-cation/sulfate-binding pocket typical of the family). A Kyte–Doolittle

hydrophobicity scan (19-residue window) yields a maximum of only 0.94 with **zero windows exceeding the 1.6 transmembrane threshold**, indicating no membrane-spanning segment — consistent with a soluble, cytoplasmic enzyme.

Domain architecture and evolutionary classification (bioinformatic evidence)

InterPro/Pfam/TIGRFAM analysis of O69787 (512 aa) reveals a two-module enzyme:

Module	Database IDs	Residues	Role
Catalytic sulfatase domain	PF00884 / IPR000917, in alkaline-phosphatase-like core superfamily IPR017850	~6–352 (core 5–486)	FGly/metal catalytic machinery
Sulfatase signature 1	PS00523 / IPR024607	52–64	Contains catalytic **Cys54→formylglycine**
Sulfatase signature 2	PS00149 / IPR024607	94–104	Second active-site residue (104)
Choline-sulfatase C-terminal domain	PF12411 / IPR025863	457–508	Choline-sulfatase-specific; candidate specificity/structural determinant

Critically, the whole protein is assigned to **dedicated choline-sulfatase families — InterPro IPR017785, TIGRFAM TIGR03417 ("choline-sulfatase", an equivalog family whose members share the same specific function), and NCBI CDD cd16032** — not merely to the generic arylsulfatase class. This provides **evolution/orthology-based evidence** that BetC's specific physiological substrate is choline-O-sulfate, consistent with the direct enzymatic assays. The dedicated C-terminal domain (res 457–508), present in choline-sulfatases but absent from generic arylsulfatases, is a plausible determinant for recognition of the bulky quaternary-ammonium (choline) moiety.

3. Pathway Context — the *betICBA* operon

BetC is the committed entry enzyme of the glycine betaine biosynthesis/catabolism pathway:

Gene	Product	Step
betI	TetR-family repressor	Regulation (choline-responsive)
betC	**Choline-O-sulfatase**	**choline-O-sulfate → choline + sulfate**
betB	Betaine-aldehyde dehydrogenase	betaine aldehyde → glycine betaine
betA	Choline dehydrogenase	choline → betaine aldehyde

Pathway: **choline-O-sulfate →(BetC)→ choline →(BetA)→ betaine aldehyde →(BetB)→ glycine betaine.**

- The four genes "constitute a single operon," comprising regulator *betI* plus structural *betC*, *betB*, *betA* (Mandon et al. 2003, P 12906115; Osterås et al. 1998, P 9736747).
- The operon is organized with 130-bp mosaic elements between *betI–betC* and *betB–betA* (P 9736747).

Regulation

Transcription shows "strong induction by choline at concentrations as low as 25 µM and to a lesser extent by choline-O-sulfate and acetylcholine but not by osmotic stress or oxygen" (Mandon et al. 2003, P 12906115). BetI is a repressor that senses the endogenous choline pool; repression is relieved when choline accumulates. Importantly, *betC* is **substrate-induced, not osmotically induced** — consistent with a metabolic entry role rather than a direct osmotic-stress gene.

4. Biological Role and Physiological Significance

1. **Osmoprotection (indirectly).** The choline liberated by BetC feeds glycine betaine synthesis. Glycine betaine is "a potent osmoprotectant accumulated by *S. meliloti* to cope with osmotic stress" (P 9736747). Notably, in *S. meliloti* choline-O-sulfate is **not** used as an osmoprotectant per se; it must be metabolized by BetC into choline and then glycine betaine (P 9736747).
2. **Nutrient catabolism (C, N, and S).** *S. meliloti* "can use this compound [choline-O-sulfate] as the sole carbon, nitrogen, and sulfur source for growth and that depends on a functional *bet* locus" (P 9736747). BetC specifically liberates the **sulfate** (sulfur source) by cleaving the sulfate ester, and unlocks the choline carbon/nitrogen skeleton for downstream catabolism.
3. **Ecological/symbiotic relevance.** Choline-O-sulfate and phosphorylcholine "are found in higher plants and fungi" (P 9736747); BetC lets the rhizobium scavenge these host/soil-derived metabolites. *bet* genes (including *betC*) are expressed "within the infection threads, in young and in mature nodules" during symbiosis with *Medicago sativa* (P 12906115), indicating the pathway operates in planta where plant-derived choline compounds are available.

Transport context (how the substrate reaches BetC)

BetC acts on intracellular choline-O-sulfate; *S. meliloti* uses dedicated transporters to supply the relevant quaternary-ammonium metabolites to the cytoplasm:

- **ChoXWV** — a high-affinity, highly specific **choline** ABC transporter (ChoX binding protein K_D ≈ 2.7 µM for choline; low affinity for acetylcholine; does **not** bind betaines). It is induced by choline (not salt) and is present in *Medicago sativa* bacteroids (Dupont et al. 2004, P 15342567).
  This supplies free choline (either imported directly or liberated by BetC) to the BetA/BetB steps.
- **BetS** — a Na^+-driven BCCT-family high-affinity **glycine betaine/proline betaine** transporter (K_m ≈ 16 µM for glycine betaine) required for early osmotic adjustment; choline is not a BetS substrate (Boscari et al. 2002, P 11976294).

Thus imported **choline-O-sulfate is hydrolyzed in the cytoplasm by BetC** to choline + sulfate; choline (from BetC or from ChoXWV) is oxidized to glycine betaine, which can additionally be scavenged directly via BetS. All of these components are choline-inducible and bacteroid-expressed, defining a coordinated choline/choline-O-sulfate catabolic and osmoadaptive module.

5. Subcellular Localization

- **Cytoplasmic / soluble enzyme.** Choline sulfatase activity was assayed in cell extracts, and choline-O-sulfate is taken up and metabolized intracellularly in *S. meliloti* (P 9736747).
  Water-soluble, cytosolic localization is the norm for non-membrane sulfatases of this family (e.g., human ARSA/ARSB and bacterial PAS are water-soluble enzymes; only steroid sulfatase carries a membrane-anchoring hydrophobic domain — Ghosh 2007, P 17558559).
  Sequence analysis of BetC confirms no transmembrane helix (max 19-residue Kyte–Doolittle hydrophobicity 0.94; 0 windows above the 1.6 TM threshold) and no strongly

hydrophobic cleavable N-terminal signal peptide, supporting soluble, cytoplasmic localization consistent with intracellular metabolism of transported choline-O-sulfate.

- The downstream glycine betaine biosynthetic steps (BetA/BetB) and glycine betaine accumulation occur in the cytoplasm, consistent with BetC acting cytoplasmically as the first step.

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6. Comparative Context (orthologs/analogs)

- ***Bacillus subtilis*** and ***S. meliloti***: *betC* is linked to osmoprotection (choline-O-sulfate feeds glycine betaine) (P 9736747 ; P 17116241).
- ***Pseudomonas putida* KT2440**: *betC* (choline sulfatase) lies next to an ABC transporter and a LysR regulator, separate from *betBA*; there a *betC* deletion "still accumulated intact COS but failed to use this compound as carbon or nitrogen source," and *betC* "was downregulated at high salt concentrations, showing that the principal role of this gene lied in COS metabolism, not in osmoprotection" (Galvão et al. 2006, P 17116241). This highlights that the **conserved biochemical function (choline-O-sulfate hydrolysis) is the same**, but the physiological coupling to osmoprotection vs. pure catabolism differs by organism. In *S. meliloti*, BetC is coupled to both.

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7. Supported vs. Refuted Hypotheses

Supported: - H1: BetC is a choline-O-sulfatase hydrolyzing choline-O-sulfate → choline + sulfate. ✔ (direct enzymatic + genetic evidence, P 9736747) - H2: BetC also hydrolyzes phosphorylcholine, at lower rate. ✔ (P 9736747)

- H3: BetC is the entry enzyme of the *betICBA* glycine-betaine pathway and is choline-inducible via BetI. ✔ (P 12906115)
- H4: BetC enables use of choline-O-sulfate as a sulfur (and C/N) source. ✔ (P 9736747)
- H5: BetC uses the conserved formylglycine-dependent sulfatase mechanism. ✔ (strong family/structural inference, P 11435113, P 17558559)

Refuted / not supported: - That choline-O-sulfate itself serves as an osmoprotectant in *S. meliloti* — refuted; it must be converted by BetC to glycine betaine (P 9736747).
- That *betC* is induced by osmotic/oxygen stress — not supported; induction is by choline, not osmotic stress (P 12906115).

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8. Limitations and Future Directions

- No experimental 3D structure or kinetic constants (Km/kcat) for *S. meliloti* BetC are available; the catalytic residue/mechanism is inferred from family homologs (PAS, human ARSA/ARSB). AlphaFold modeling and direct identification of the FGly-bearing cysteine and metal site would confirm the mechanism.
- The formylglycine-generating machinery (FGE/anSME) responsible for maturing BetC in *S. meliloti* has not been explicitly linked; verifying the maturation route would complete the mechanistic picture.
- Precise subcellular localization (cytoplasm vs. periplasm) and the choline-O-sulfate uptake transporter in *S. meliloti* warrant direct experimental confirmation.

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9. Consolidated Evidence Summary

Claim about BetC	Evidence type	Key support
Choline-O-sulfatase: choline-O-sulfate → choline + sulfate (EC 3.1.6.6)	Experimental (enzyme assay)	Osterås 1998 (P 9736747)
Activity assigned to *betC* gene	Experimental (mutant: activity lost in *betC*, retained in *betB*)	P 9736747
Also hydrolyzes phosphorylcholine (lower rate)	Experimental	P 9736747
Enables use of choline-O-sulfate as sole C/N/S source	Experimental (growth, *bet*-dependent)	P 9736747
Entry enzyme of *betICBA* operon; choline-inducible via BetI; bacteroid-expressed	Experimental (operon mapping, *lacZ* fusions, in planta)	Mandon 2003 (P 12906115)
Formylglycine-dependent catalytic mechanism (FGly hydrate nucleophile + metal)	Structural (family homologs)	Boltes 2001 (P 11435113); Ghosh 2007 (P 17558559)
Catalytic Cys54→formylglycine; 2nd active site 104; metal site; no TM helix	Bioinformatic (sequence + UniProt features + hydrophobicity)	This work (O69787 analysis)
Dedicated choline-sulfatase equivalog family; 2-domain architecture	Evolution/bioinformatic (InterPro/TIGR03417/ cd16032)	This work
Cytoplasmic; substrate delivered by ChoXWV (choline) / BetS (betaine)	Experimental + inference	Dupont 2004 (P 15342567); Boscari 2002 (P 11976294)

References (PMIDs)

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  P 9736747

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  P 12906115

- Galvão TC, de Lorenzo V, Cánovas D. *Uncoupling of choline-O-sulphate utilization from osmoprotection in Pseudomonas putida.* 2006.

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  P 15342567

- Boscari A, Mandon K, Dupont L, Poggi MC, Le Rudulier D. *BetS is a major glycine betaine/proline betaine transporter required for early osmotic adjustment in Sinorhizobium meliloti.* J Bacteriol 2002.

  P 11976294